DEPI INDUSTRY DELIVERABLE — MODULE 1

Project Planning & Management

Project: Automated UI Testing for an E-Commerce Website

Team Members & Responsibilities
Leader: Mohammed Ahmed Atta Zyad Yasser Omar Hafsa Nabil Yahia Hanaa Mohamed Okla
Mariam Mohamed Farrag

1. Project Proposal & Overview

Objective: Develop a clean, maintainable, and robust automated UI test suite for the primary end-to-end user journeys of a public e-commerce platform (e.g., Sauce Labs "Swag Labs" application).
Scope: Automated verification of core critical flows including user authentication, product search and cart management, and end-to-end checkout execution using Selenium WebDriver and the Page Object Model (POM) design pattern.
Technology Stack: Selenium WebDriver, Python / Java, PyTest / TestNG, Git & GitHub, Page Object Model (POM) Architecture.

2. Task Assignment & Project Plan (Timeline)

PHASE / WEEK	TASKS DESCRIPTION	ASSIGNED TO	KEY DELIVERABLES
Week 1: Setup & Baseline	Create folder structure (Mohammed); Set up local environment (Selenium/ChromeDriver) & select target site (Zyad, Hafsa); Write baseline homepage title check test (Hanaa, Mariam).	Mohammed, Zyad, Hafsa, Hanaa, Mariam	Working local test environment, baseline check script, target site document.
Week 2: Core Journeys	Develop manual test cases for login & cart (Hafsa); Automate user login journey (Zyad); Automate "add to cart" journey (Hanaa); Implement dynamic explicit waits (Mariam, Mohammed).	Hafsa, Zyad, Hanaa, Mariam, Mohammed	Automated login script, add-to-cart script, wait-stabilized test execution.
Week 3: Checkout & Verification	Automate checkout process and dummy form submission (Mariam); Enhance all scripts with detailed assertions (Hanaa, Zyad); Externalize test data management (Mohammed).	Mariam, Hanaa, Zyad, Mohammed	Automated checkout flow script, assertion-enhanced scripts, structured test data.
Week 4: POM & Final Summary	Refactor code into Page Object Model (Zyad, Mariam); Organize and execute full suite (Hafsa, Hanaa); Author final summary report (Mohammed).	Zyad, Mariam, Hafsa, Hanaa, Mohammed	POM-refactored codebase, final execution report, documented project repository.

3. Risk Assessment & Mitigation Plan

IDENTIFIED RISK	IMPACT	LIKELIHOOD	MITIGATION STRATEGY
Flaky Tests (Dynamic UI Loading)	High	Medium	Enforce explicit dynamic waits (WebDriverWait) instead of static delay timers.
UI Locator Changes in Application	High	High	Implement Page Object Model (POM) to centralize locators and reduce code updates.
Browser Driver & Environment Mismatches	Medium	Medium	Standardize driver management across team machines using WebDriverManager.
Unstable or Invalid Test Credentials	Medium	Low	Externalize synthetic test accounts and customer details into separate data files.

4. Key Performance Indicators (KPIs)

Test Suite Pass Rate: Minimum 95% pass rate across all automated end-to-end execution runs.

Test Coverage: 100% automated coverage of identified core user journeys (Login, Search, Cart, Checkout).

Code Maintainability: Zero duplicate element locators across test scripts via strict POM architecture.

Execution Reliability: Less than 2% test failure rate caused by timing or synchronization delays.